School Worksheet

Class 12 Mathematics, Chapter 9: Differential Equations

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The order and degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^5 = \frac{d^2y}{dx^2}$ are:

(A) order 1, degree 1 **(B)** order 1, degree 2

(C) order 2, degree 1 **(D)** order 2, degree 2

2 MCQ — 1 mark

The degree of the differential equation $\frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$ is:

(A) 1 **(B)** 2 **(C)** 0 **(D)** not defined

3 MCQ — 1 mark

The integrating factor of the differential equation $x\frac{dy}{dx} - y = x^2$ is:

(A) x **(B)** $-x$ **(C)** $\frac{1}{x}$ **(D)** $\log x$

4 MCQ — 1 mark

The number of arbitrary constants in the general solution of a differential equation of order 4 is:

(A) 0 **(B)** 2 **(C)** 3 **(D)** 4

5 MCQ — 1 mark

The general solution of $\frac{dy}{dx} = e^{x+y}$ is:

(A) $e^x + e^{-y} = C$ **(B)** $e^x + e^y = C$

(C) $e^{-x} + e^y = C$ **(D)** $e^{-x} + e^{-y} = C$

Q No.	Question	Marks
6	Assertion-Reason **Assertion (A):** $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 4y = 0$. **Reason (R):** A function is a solution of a differential equation if it satisfies the equation identically. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find the general solution of $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.	2 marks

Level 2

18 marks

8	Short answer A bacteria sample grows exponentially. The differential equation representing the growth is $\frac{dP}{dt} = kP$, where P is the population at time t. Obtain the general solution and express P as an exponential function of t.	2 marks
9	Short answer Find the particular solution of $\frac{dy}{dx} = y\tan x$, given that $y = 1$ when $x = 0$.	3 marks
10	Short answer Solve: $x\frac{dy}{dx} + 2y = x^2$ $(x \neq 0)$.	3 marks
11	Short answer Solve the homogeneous differential equation $x\frac{dy}{dx} = x + y$.	3 marks
12	Short answer Verify that $y = Ax + \frac{B}{x}$ is a solution of $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$.	2 marks
13	Short answer Find the general solution of $\frac{dy}{dx} = \sin^{-1} x$.	2 marks
14	Short answer Find the particular solution of the differential equation $\frac{dy}{dx} - 2xy = 3x^2e^{x^2}$, given $y(0) = 5$.	3 marks

Q No.	Question	Marks
Level 3		**25 marks**
15	Long answer Solve: $x\,dy - y\,dx = \sqrt{x^2 + y^2}\,dx$.	5 marks
16	Short answer Find the particular solution of $(1 + x^2)\dfrac{dy}{dx} + 2xy = \dfrac{1}{1 + x^2}$, given $y = 0$ when $x = 1$.	3 marks
17	Long answer Find the particular solution of $x \sin\left(\dfrac{y}{x}\right)\dfrac{dy}{dx} + x - y\sin\left(\dfrac{y}{x}\right) = 0$, given $y = \dfrac{\pi}{2}$ when $x = 1$.	5 marks
18	Short answer Solve the differential equation: $x^2\,dy + y(x + y)\,dx = 0$.	3 marks
19	Long answer Solve: $(\tan^{-1} y - x)\,dy = (1 + y^2)\,dx$.	5 marks
20	Case study **Case Study:** In a laboratory culture, the number of bacteria P increases at a rate proportional to the number present. The count of 1,00,000 bacteria increases by 10% in 2 hours. **(i)** Write the differential equation modelling this situation and state its order and degree. — 1 mark **(ii)** Solve it to express P in terms of t and find the constant k. — 1 mark **(iii)** In how many hours will the count reach 2,00,000? — 2 marks	4 marks

Answer key

Chapter 9: Differential Equations, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(C) order 2, degree 1
2	(D) not defined
3	(C) $\dfrac{1}{x}$
4	(D) 4
5	(A) $e^{x} + e^{-y} = C$
6	(A) Both A and R are true and R is the correct explanation of A.
7	$\tan^{-1} y = \tan^{-1} x + C$

Level 2

8	$P = P_0 e^{kt}$, where P_0 is the population at $t = 0$
9	$y = \sec x$
10	$y = \dfrac{x^2}{4} + \dfrac{C}{x^2}$
11	$y = x \log \lvert x \rvert + Cx$
12	LHS $= 0 =$ RHS, so $y = Ax + \dfrac{B}{x}$ is a solution.
13	$y = x \sin^{-1} x + \sqrt{1 - x^2} + C$
14	$y = (x^3 + 5)\, e^{x^2}$

Level 3

15	$y + \sqrt{x^2 + y^2} = Cx^2$
16	$y(1 + x^2) = \tan^{-1} x - \dfrac{\pi}{4}$
17	$\cos\left(\dfrac{y}{x}\right) = \log \lvert x \rvert$
18	$x^2 y = C(y + 2x)$
19	$x = \tan^{-1} y - 1 + Ce^{-\tan^{-1} y}$
20	(i) $\dfrac{dP}{dt} = kP$, order 1, degree 1; (ii) $P = 100000\, e^{kt}$, $k = \dfrac{1}{2}\log 1.1$; (iii) $t = \dfrac{2 \log 2}{\log 1.1} \approx 14.5$ hours